

Encode

Shuffle

Analyze

Trusted Setup



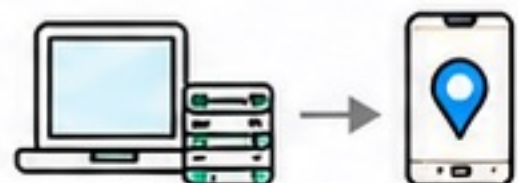
Groth16 setup for TSIPMAIN

- pk → Clients
- vk → Shuffler



Enrollment Attestation Authority (EA)

- ✓ federated enrollment anchors
- ✓ threshold attestation (t-of-n)
- ✓ attested enrollment metadata
- ✓ tier_vmax_sq
- ✓ modeet_commitment
- ✓ secret_commitment



Enrollment / Warmup (off the aggregation path)

- warmup updates state
- not aggregated
- $N_{warm} = \max(K, 2)$

- = data flow
- - -> = trust / assumption
- ✓ = accepted
- ✗ = rejected

Stage 1: Client-Side Processing



1. Local trajectory observations

- hidden coordinates
- timestamp
- previous / anchor commitments



2. Payload binders & ciphertext package

$$primary_com = \text{Poseidon}(primary, \sigma_{pay}, r_{pri})$$

$$share_content_com = \text{Poseidon}(primary_com, \sigma_{share}, r_{share})$$

$$blob_hash = H(blob_A, blob_R, ctx)$$

- $primary_com$: hidden-primary binder
- $share_content_com$: plaintext-share-content binder
- $blob_hash$: transport digest only



3. Commitment chain update

- Poseidon location commitments
- MMC chain
- binds state, round, and payload binders

4. Proof generation

- ✓ per-step continuity
- ✓ ADWC window continuity
- ✓ hidden-cell encoding
- ✓ binder consistency
- ✓ secret / mode binding

Π_{val} proof

- ✓ same $primary_com$
- ✓ same $share_content_com$
- ✓ same $blob_hash$
- ✓ share package valid for committed primary

report = TSIPMAIN + Π_{val} +
binders + ciphertext package

Stage 2: Shuffler-Side Verification

- 1 chain-state + timestamp monotonicity
predecessor commitment ✓
- 2 public-signal consistency
agn, anchor, mode ✓
- 3 recompute blob_hash
ciphertext-package transport check ✓
- 4 verify TSIPMAIN
trajectory-to-binder consistency ✓
- 5 verify Π_{val}
binder-to-share semantic validity ✓
- 6 state update / hold / blacklist
replay / identity / drift ⊖

check shared binders:
 $primary_com, share_content_com, blob_hash$

invalid updates are rejected
before secure aggregation



Payload swap (A6) is rejected if
 $primary_com / share_content_com /$
 $blob_hash$ mismatch or Π_{val} fails.

Verification & Security Semantics



continuity
soundness



zero-
knowledge



payload
semantic binding



proof-carrying
integrity filter



public-output
DP

TSIPMAIN: trajectory-
to-binder consistency

Π_{val} : binder-to-share
semantic validity

$blob_hash$: transport
immutability only

forwarded
accepted
reports

Stage 3: Dual Secure Aggregation & DP Release

Aggregation

Aggregator A

- decrypt blob_A
- re-check format / context
- add valid share_A

A and R
do not collude

Aggregator R

- decrypt blob_R
- re-check format / context
- add valid share_R



Decoder:
reconstruct
aggregate

DP Release



SVT:
threshold
filtering



Laplace:
perturbation



Published
DP heatmap

- public-output DP applies only to the released heatmap; A and R must not collude.
- TSIP integrity is enforced upstream of aggregation.